HATCHSIDE · DESIGN DECISIONS REGISTER

HatchSide design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-11-05	Amish Chadha	Kit 1.9.1 upgrade; open decisions D13 to D17 added (proposed, awaiting Amish); STEM check recorded
0.1	2026-10-03	Amish Chadha	Register opened at TRL 3; every decision made under Amish's 2026-10-03 pre-approval

HatchSide design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

> **Safety:** Several decisions below set the safety case of a lifting frame used over confined spaces. Each took the conservative option and names the evidence that would relax it. HatchSide is a material-handling frame for tool heads; it is never used to lift or lower a person, and nobody enters the space.

Open decisions

All proposed at the kit 1.9.1 upgrade (2026-11-05), awaiting Amish. Nothing below is decided.

#	Decision	Recommendation	Options	Record
D13	Concept: tripod and hand winch (A) or straddle-frame davit (B)	**Proposed, awaiting Amish:** A, which scores 160 of 180 against 141 for B (13.8 % lead, outside the 10 % band)	A; B; keep both open	HTS-OPT-001; HTS-DDR-003
D14	Confirm the corrections made when the model was rebuilt to D3: one shackle per chain lug, a 44 x 45 mm lug on a 140 x 360 mm foot plate (80 mm longer), a 14.2 mm longer head interface, M16 x 110 hinge bolts, and the model fixes K1 to K11	**Proposed, awaiting Amish:** confirm; no price changes and no change to the pitch	Confirm; revert any item (the D3 check then fails until it is remodelled)	HTS-DDR-004
D15	Offer a second, lower winch bracket position on leg A, the idea taken from option B (winch at waist height)	**Proposed, awaiting Amish:** not applied; test with a crew first	Apply to the model and BOM; leave as is	HTS-OPT-001
D16	The grab nets 9.7 L against the 10 L of PR-008 once the 10 % drip is counted (STM S9)	**Proposed, awaiting Amish:** lengthen both shells by about 25 mm (gross 16.4 L, net 10.3 L); the plan diagonal becomes 542 mm against the 540 mm limit, so either accept 2 mm less spare in a 560 mm opening or narrow the shell plates; not applied to the model	Lengthen the shells; keep the grab and restate PR-008 as 9 L net; raise the fill factor after a trial	HTS-STM-001 S9; HTS-CAL-001 section 8
D17	The hand winch can pull its 450 kg rating, so a snag could load the rope to 4.4 kN (factor 4.8) and leg A to 6.2 kN (STM S24)	**Proposed, awaiting Amish:** add a shear link or friction clutch rated at about 1.5 times the safe working load in the crank drive; not applied	Add a limiter; accept the stall case as it is; choose a winch rated nearer 300 kg	HTS-STM-001 S24

Standing instruction (2026-10-23): STEM-check fixes that keep the working principle are implemented and recorded here as decided under that instruction. The STEM check (HTS-STM-001) found no failing claim, so no such fix was made.

To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The winch's base hole pattern	Sets the four winch holes in the front bracket plate	HTS-DDR-002, C5
2	The winch drum holds 15 m of 6 mm rope with three turns to spare, its first-layer rating is at least 450 kg, and its drum core is large enough for 7x19 rope (the model's core is 60 mm, D/d about 10)	R3 and R4 rest on these catalogue figures	HTS-CAL-001, A8
3	The rope's certificate shows a minimum breaking load of at least 21 kN	Rope safety factor of 14.3	HTS-CAL-001, A7
4	The chain's working load limit is at least 5 kN, and 2.78 m plus the three lug shackles sets the 3.0 m foot circle	Chain tension 1,195 N at the design case; leg spread	HTS-CAL-001, section 3
5	The sheave's bore is 20 mm, its width fits between cheeks 34 mm apart with the two washers, and its rating is at least 500 kg	Axle and cheek spacing	HTS-DDR-002, C3
6	The ball-lock pins' 70 mm grip suits the 60 mm upper leg	Leg adjustment	HTS-DDR-002, C1
7	The EN 362 connector's gate opening fits the rings of common rescue lines	Retrieval head use	HTS-DDR-001, D2
8	A generic roller-applied digester sealant is sold in the first region, and its cure needs no entry	DomeReach	HTS-DDR-001, D8

Value engineering

Value-engineering target: USD 5,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 2,838 (USD 2,162 under the target). Main cost drivers and savings worth trying:

- The largest lines are the hand winch (USD 320), the grab head (USD 260), the rim guard panels (6 x USD 48), the DomeReach head (USD 225), the sampler (USD 145) and the tripod head (USD 140).
- Galvanizing is priced at a minimum lot for each steel part; sending all steel parts in one lot is the simplest saving.
- The rim guard could use HDPE board on a steel tube frame from local stock, or reuse traffic barrier panels, if cost matters more than weight.
- Making the design constructable added the foot chain lugs and longer foot plates (USD 6 across three feet) and longer chains (USD 12).

Decisions made

All decided by Amish under his pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost."

Date	Decision	Decided by	Record
2026-10-03	Safe working load 150 kg for tool heads, proof 1.5 times; WellSling does not share the frame (D1)	Amish, pre-approval quoted above	HTS-DDR-001
2026-10-03	Retrieval head kept as a clip-on guide only; the haul is made with certified rescue equipment. Conservative safety option; relaxing it needs personnel certification of frame, winch and rope (D2)	Amish, pre-approval quoted above	HTS-DDR-001
2026-10-03	Two-speed hand winch with an automatic load brake, 450 kg (D3); 15 m of 6 mm 7x19 grade 316 rope (D4)	Amish, pre-approval quoted above	HTS-DDR-001
2026-10-03	6082-T6 square aluminium legs, telescoping (D5); hot-dip galvanized S275 and A4 stainless (D6)	Amish, pre-approval quoted above	HTS-DDR-001
2026-10-03	DomeReach: 2.5 m design depth (D7), generic sealant (D8), single hinged arm and no gripper (D9)	Amish, pre-approval quoted above	HTS-DDR-001
2026-10-03	Requirements R13 (heads pass 560 mm) and R14 (stable at 10 degrees) added (D10)	Amish, pre-approval quoted above	HTS-DDR-001
2026-10-03	India first; first candidates to approach, not agreed: a municipal sewer department running mechanised cleaning, and a household biogas programme in India or Nepal (D11)	Amish, pre-approval quoted above	HTS-DDR-001
2026-10-03	Value-engineering target stays USD 5,000; overruns accepted; `budget_usd` unchanged (D12)	Amish, pre-approval quoted above	HTS-DDR-001
2026-10-03	Design for construction C1 to C12: telescoping square legs, sleeved M16 joints, cheek-plate mast, winch bracket with stop bolt, clevis feet with outboard chain lugs, standard 10 mm tab, closing grab, pinned rim guard, guide-only retrieval head, single-hinge DomeReach, checked clearances	Amish, pre-approval quoted above	HTS-DDR-002
2026-10-03	Appearance model adds only labels (safe working load, "never lift a person"), a street with a manhole and a 1.75 m person for scale; no change to the design	Amish, pre-approval quoted above	`docs/REVIEW.md`, 2026-10-03